CHRISTIAN OMAR PAYÁN TORRÓNTEGUI

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SUMMARY

Robotics & Digital Systems Engineering student (8th semester) with expertise in process optimization, mechanical design, and production technology. Experienced in agile methodologies, data tracking, and continuous improvement for engineering workflows. Seeking a Production Engineering Intern role to apply analytical modeling, quality control standards, and operational efficiency.

EDUCATION

B.S. Robotics and Digital Systems Graduating Expected December 2026
Tecnológico de Monterrey
Zapopan, Jalisco
Relevant coursework: Process Optimization, Industrial Mounts, Agile & Scrum Methodologies

TECHNICAL SKILLS

Office & Productivity: Microsoft Office (Excel, Word, PowerPoint, Outlook — 100-hour certification), Data Analysis
Programming & Languages: Python, C++, Spanish (Native), English (B2 Professional)
Design & Manufacturing: SOLIDWORKS, Fusion 360, Advanced FDM 3D Printing, Prototyping, Eagle
Robotics & Systems: ROS 2, Gazebo, URDF, STM32, ESP32
Methodologies & Operations: Agile, Scrum, Process Improvement, 5S Concepts, Safety & Quality Documentation

PROFESSIONAL EXPERIENCE

Industrial Mounts & Support Interfaces, Zapopan, Jalisco: Mechanical & Product Designer Feb-Jun 2026

- Engineered mechanical interfaces for controllers, developing a 3-DOF system that reduced fabrication costs by 95% vs commercial alternatives
- Designed and prototyped a custom mini PC enclosure for robust deployment in industrial environments, optimizing material selection for thermal and mechanical resistance

ACADEMIC & TECHNICAL PROJECTS

Open-Source Gastrointestinal Medical Phantom Ago-Dic 2025 – Feb-Jun 2026
Agile, Advanced FDM

- Engineered phantom via advanced FDM 3D printing (TPU/PETG), reducing prototyping costs by 80% vs commercial alternatives
- Managed development using Agile methodologies to iterate rapidly on anatomical accuracy, quality specs, and material constraints

Port Automation & Drone Sim Ago-Dic 2025
ROS 2, Gazebo, C++, Python

- Engineered URDF models and C++/Python ROS 2 nodes for kinematic control and reliable flight telemetry
- Designed & 3D-printed custom hardware: drone mirror/lens mounts, and an AMR bracket (LiDAR, ArUco, Jetson)

DigiMix Ago-Dic 2024
Embedded Systems, Scrum

- Designed custom PCB and 3D-printed casing; engineered hardware using dual-core STM32 and ESP32 (WebSockets)
- Applied Scrum to manage lifecycle, integrating hardware and software architectures with zero logic faults